

HandNav: An IoT-Based Smart Navigation Glove for Visually Impaired People

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Abstract—Visually impaired people face severe difficulties in moving around dynamic settings, frequently using conventional assistive devices that provide little awareness of the situation and predict hazards. This paper presents a IoT-enabled smart glove, low-cost and wearable assistive device that can help improve mobility, safety, and independence for the visually impaired. The proposed system combines ultrasonic detection of real-time obstacles (2–400 cm in range) and flame sensors to monitor fire hazards and GPS to track their exact location and continuously monitor them with Bluetooth. When the system identifies impediments or threats within pre-set ranges, immediate audio warnings are produced through an in-built buzzing alarm, and emergency events cause caregiver notifications based on location through a mobile application. Experimental assessment shows valid obstacle detection under one-second response times, flame detection, and GPS positioning error within 3–5 m in the outdoor conditions. The entire system is designed at a low price point, which is affordable to many people, especially in resource-limited places such as Bangladesh, where there are 1.2 million blind people. The Smart Glove will help to provide a viable, scalable solution to the problem by utilizing several safety functions in one wearable item, which will restore the confidence and foster the independent lives of the visually impaired, such as obstacle detection, flame detection, and GPS navigation to assist them in their daily activities.

Index Terms—Smart glove, visually impaired, assistive technology, Internet of Things (IoT), obstacle detection, GPS tracking, flame sensor, real-time alert system, wearable device, low-cost solution.

I. INTRODUCTION

Independent mobility is a cornerstone of human existence, and it allows individuals to engage in social, economic, and personal activities with dignity and confidence and participate in them as full citizens of a society [7]. Safe and independent navigation by the visually impaired in everyday environments is, however, a thorny issue. The loss of sight has a great impact on the capability of an individual to see in the distance, evaluate distances, identify dangers, and react to changing environments like moving traffic or walkways busy with pedestrians [2]. Blindness drastically reduces an individual's capability to view objects in the distance, judge distances, locate dangers, and respond to dynamic environments like traffic or walkways full to capacity with pedestrians, ruling out distances necessarily. This condition causes most people who are visually impaired to live on the mercy of those around

them to conduct their daily movements, and this diminishes their personal independence, leading to a negative impact on their psychological well-being and general living standards. Bangladesh alone has an estimated 1.2 million blind and over 351,000 individuals with low vision, with most having a severe case among individuals older than 50 years of age who suffer from blindness [8]. This condition brings out a significant national burden, especially to the aged population. Even though the use of traditional assistive devices like white canes and trained guide dogs offers marginal mobility assistance, it does not have a lot of environmental awareness, a limited range of obstacles, or low adaptability to complicated scenarios [12]. In addition, guide dogs are very expensive in terms of training and maintenance, and white canes have to rely on physical contact. As embedded systems, sensors, and Internet of Things (IoT) technologies are rapidly developing, it is essential to have an affordable, user-friendly, and intelligent device that will enhance real-time environmental awareness and contribute to improved safety during the navigation process to a great extent.

- **Obstacle detection:** Ultrasonic detectors sense the objects around them in the range of 2–400 cm and give feedback instantly to assist users to prevent the occurrence of collision.
- **Emergency and hazard support:** An emergency button is equipped with GPS and transmits alerts and real-time information about the user's position, and a fire sensor notifies users of fire hazards.
- **IoT-based communication:** It can be used to maintain a continuous flow of data to a mobile application, allowing it to monitor items in real-time and react swiftly to emergency scenarios using IoT-based communication.

Overall, the suggested Smart Glove is a wearable assistive device that will enhance the self-mobility and personal safety of people with visual impairments. The system combines ultrasonic sensing, location tracking with the use of GPS, emergency alerts, and detection of fire hazards into a single small unit. The Smart Glove provides a chance to exchange information and feedback between the user, caregivers, and the relevant authorities in real time, which is made possible by using IoT-enabled communication. This sensor-based solution

has a higher response time, enhanced situational awareness, and less reliance on external aid or education as compared to conventional assistive solutions. All in all, the Smart Glove is a confidence-boosting, independent, and daily safety product and a scaling and technology-based mobility aid for the visually impaired.

II. PROBLEM STATEMENT

The visually impaired individuals have their daily challenges as they move about in their environment, particularly in complex and busy places. They are more exposed to accidents because they have little capacity to easily identify any obstacles, unexpected changes, or hazardous occurrences. Even routine tasks such as walking or moving around are usually supported, and they are not as independent as they can be. Safety hazards including fire or extreme heat are also dangerous besides the physical obstacles. A lot of the visually impaired individuals cannot identify them early enough, causing serious injuries. Although some assistive technologies exist, they lack real-time hazard detection and emergency assistance features, such as location tracking.

In addition, the majority of the assistive devices that are available are costly, complicated to operate, or not readily available for all. This brings about the requirement of a low-price, straightforward, and dependable wearable item that can offer real-time obstacle detection, security warnings, and positioning. We can effectively address this issue to improve the mobility, safety, and living standards of the visually challenged.

III. LITERATURE REVIEW

Over the past years, there have been many researches conducted on assistive technologies in order to help individuals with poor eye sight or disabled. The majority of those are focused on obstacle detection, navigation aid, real time monitoring, and communication aid based on various sensors, microcontrollers and wireless technologies. The main aim of these systems is to improve the safety, mobility, independence and quality of life among the users on daily setting. Hummadi et al. have introduced an assistive navigation system based on IoT, according to which the obstacles that were present nearby were detected in real-time and recorded by ultrasonic sensors connected to an Arduino microcontroller [2]. The system offers instant vibration or sound warning when an obstacle is detected so that users can take corrective measures. The suggested low cost and energy saving design is exceptional in enhancing the safety of the user, space sense and self-reliance. According to the study by Vijay et al. [3], an IoT-based smart glove that would support communication with deaf and mute people was developed. The system uses several flex sensors to record the movements of the fingers and then they are interpreted by an Arduino UNO microcontroller. The sensor data is still sent through an IoT server with the help of an ESP8266 Wi-Fi module and the gesture patterns are compared to a database of known patterns to produce audio output. The

TABLE I: Summary of Related Smart Glove-Based Assistive Systems

Study	Methods	Key Features	Limitations
Hummadi et al. [2]	Arduino, Ultrasonic	Real-time obstacle alerts	No GPS/emergency
Vijay et al. [3]	Arduino, Flex, ESP8266	Gesture communication	No navigation
Latif et al. [1]	Ultrasonic knee glove	Low-height obstacle detection	No GPS/AI
Sai Sumanjali et al. [13]	Ultrasonic, Buzzer, LED	Multimodal alerts	No wireless alert
Ehelagasthenna et al. [4]	Haptic E-yarn, Vibrotactile feedback	Assistive haptic communication	No obstacle detection/navigation
Beri et al. [5]	Camera, IMU, CNN	High accuracy recognition	High cost/power consumption
Senthil Kumar et al. [14]	Flex, Arduino, GSM	Gesture + emergency alert	Complex hardware design
Ketcham & Ganokratanaa [10]	Multi-sensor, GPS	Obstacle + location sharing	High system complexity
Raj et al. [11]	Arduino, GSM	Healthcare monitoring	Not navigation-based
Rozali et al. [16]	Ultrasonic, Buzzer	Distance-based alerts	No GPS/IoT integration
Proposed System	Ultrasonic, Flame, GPS, BT	Obstacle + fire alert	No vision-based AI

research proved that it has increased communication effectiveness and lowered response time as opposed to traditional systems.

An IoT-based smart glove monitoring system has been suggested by Raj et al. [11] to patients with physical challenges and speech impairment in a medical setting. The system enables the patient to inform about the needs with the help of the hand gestures, which are recognized by the gravity sensors, but the heartbeat sensor helps to track the physiological parameters, including heart rate. An Arduino Uno is used to process sensor data and wirelessly send it to a receiver that is built around an ESP32 microcontroller using an HC-12 module. On an LCD screen, the receiver presents the requests of the patient, and the message is sent to the family members and doctors through a GSM module. Moreover, the patient data is uploaded into a cloud system to ensure continuous monitoring and the analysis of this data in the future, enhancing the automation of healthcare and the efficiency of the response.

Researchers [15], [16] have suggested smart glove system as an electronic travel companion that can help blind people to avoid obstacles on their journey and guide them. The system combines ultrasonic sensors, vibration motor, and buzzer with an Arduino ATmega328 microcontroller to give real time feedback. The closer the obstacle, the higher the level of vibration and frequency of the buzzer, the more alert the user is. The experimental findings were based on the mobility and distance-related tests illustrating that the suggested smart glove is a viable tool at enhancing the perception of the environment and a supplemental tool to the conventional mobility assistants, namely white canes and guide dogs.

IV. METHODOLOGY

The proposed system is created in the form of a smart glove that is designed to fit in the form of a wearable and help visually impaired people identify their environment and have warnings about safety in a timely manner. The methodology is based on an IoT approach in which various sensors, communication modules and alert mechanisms are packaged into a small, single wearable unit. The design is aimed at simplicity, reliability, cheapness and ease of use, so that the user of the system does not have to technically understand it in order to adopt it in the routine activities.

TABLE II: Hardware Components and Purpose

Hardware	Purpose
Arduino UNO	Controls the system and processes sensor data
Flame Sensor	Detects fire or heat hazards
Emergency Button	Sends instant alerts in emergencies
Ultrasonic Sensor (HC-SR04)	Detects obstacles for safe navigation
GPS Module	Tracks real-time location
Bluetooth Module	Enables communication with a mobile app
Buzzer	Provides audible alerts for obstacles or hazards
Power Supply	Powers all components reliably

The system utilizes an ultrasonic sensor as a leading obstacle detection system. This sensor constantly sends ultrasonic waves and records the duration that the reflected signal takes, enabling the system to estimate the distance of the objects around. The distance information is continuously observed, and once an object is spotted within a set safe distance, the sensor transmits the data to the controller to be evaluated immediately. The process allows early identifying the stumbling blocks and avoidance of collisions.

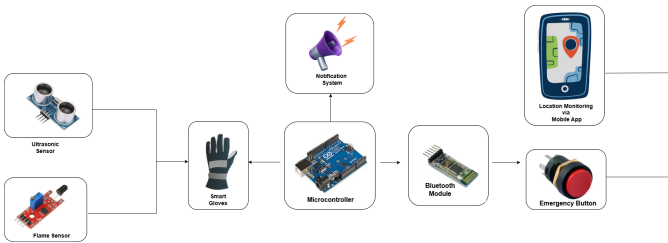


Fig. 1: Overall Methodology of the Proposed Smart Glove System

In order to overcome the safety threat posed by fire and heat, the smart glove is fitted with a flame sensor. The fire sensor continuously checks the surroundings to ascertain the presence of fire or high temperatures. When this danger is identified, the sensor device produces a signal immediately sent to the controller. This will enable the system to react promptly and give timely alerts that will enable the user to evade potentially dangerous situations. The core processing unit of the system is a microcontroller, such as an Arduino Uno or NodeMCU.

Algorithm 1 Smart Glove System Operation

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1: Initialize sensors  $S_o, S_f, U, G$ , button  $E_b$ , Bluetooth  $B$ 
2:  $A_b \leftarrow 0$  ▷ Buzzer OFF
3: while  $t \in \mathbb{R}^+$  do
4:   if  $E_b(t) = 1$  then
5:      $A_b \leftarrow 1$ ; Send_msg("Emergency") via  $B$ 
6:   end if
7:   if  $S_f(t) = 1$  then
8:      $A_b \leftarrow 1$ ; Send_msg("Flame detected")
9:   end if
10:   $d(t) \leftarrow U(t)$ ,  $L(t) \leftarrow G(t)$ 
11:  if  $0 < d(t) < \theta_d$  then
12:     $A_b \leftarrow 1$ ; Transmit  $L(t)$  via  $B$ 
13:  end if
14: end while

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The controller receives input signals of all connected sensors, works with the data based on the predetermined logic and threshold values and decides on the corresponding response of the system. This is guaranteed by the use of a micro-controller which ensures rapid decision making and effective coordination between the sensing, alerting and communication modules.

In an audio-based alert system, a buzzer is used to give the user immediate feedback. The controller detects the obstacle or hazard, and a buzzer is on to produce a warning sound. Various types of threats can be identified by using different alert patterns or varied alert durations, including hazards in the immediate vicinity (i.e. obstacles or fire hazard). This method enables the user to perceive the situation without being required to have visual feedback.

GPS and Bluetooth modules are added in the system to enable tracking and communication. The GPS module keeps on-demanding the real-time location position of the user, and it could be very crucial in emergency situations or when they need any help. The Bluetooth module can be used to provide wireless communication between the smart glove and an application installed on a mobile device, ensuring that the user with a smart glove is tracked by a caregiver or a loved one when required.

One of the aspects of the system design is power management. The smart glove has a rechargeable battery that has been chosen due to its low power consumption and long time of use. Efficient power utilization methods are also used so that the system does not need to be recharged frequently; hence, it is suitable to be used on a daily basis. The figure shows the concept of a smart assistive glove that visually impaired people work with. The smart glove is worn by the user and has sensors to identify the objects around it and possible dangers. When an object is identified, the system either gives tactile feedback or alerts the user to avoid obstacles. The glove is also fitted with a flame sensor to identify the presence of fire and to give instant warnings to promote safety. The glove is linked to a smartphone by Bluetooth to track the location and monitor it through GPS. Using wireless networks and the internet,



Fig. 2: Embedded Assistive Glove with Real-Time Hazard Detection

the system facilitates communication and real-time support, therefore making navigation safer and more independent for visually impaired people.

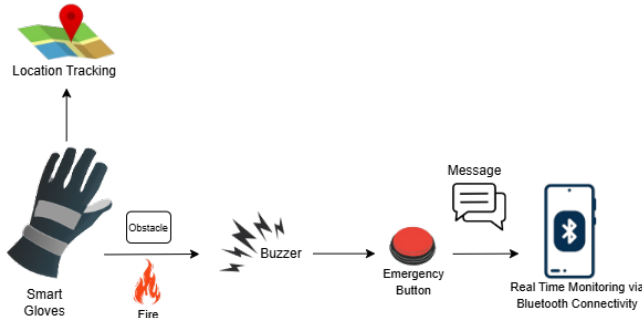


Fig. 3: System Architecture of the Smart Glove

The Arduino IDE is used to program, test, and implement the system through which the code is written, compiled and uploaded to the microcontroller. The entire sensor integration, data processing and alert logic are built and tested out directly at the hardware level through this platform. The Arduino IDE has made the process easy, reliable, and simple to change through modification. The method helps in effective system implementation at a low cost with stable performance. On the whole, the selected methodology will help to ensure that the proposed smart glove functions properly and helps to enhance the safety, mobility, and independence of blind people.

V. EXPERIMENTAL RESULTS AND PERFORMANCE ANALYSIS

The proposed smart glove was implemented and experimentally evaluated to assess obstacle detection, alert response, safety hazard detection, and location tracking performance. The distance between the user and a nearby obstacle is calculated using the time-of-flight principle of ultrasonic waves. The distance is computed as:

$$d = \frac{v \times t}{2} \quad (1)$$

where d is the distance to the obstacle, v is the speed of sound in air (approximately 343 m/s), and t is the time taken by the ultrasonic pulse to travel to the obstacle and return.

During testing, the ultrasonic sensor reliably detected nearby obstacles within the predefined range. Alerts were generated when objects entered the warning zone, allowing sufficient reaction time. Minor variations were observed outdoors due to environmental interference. According to Figure 4, the

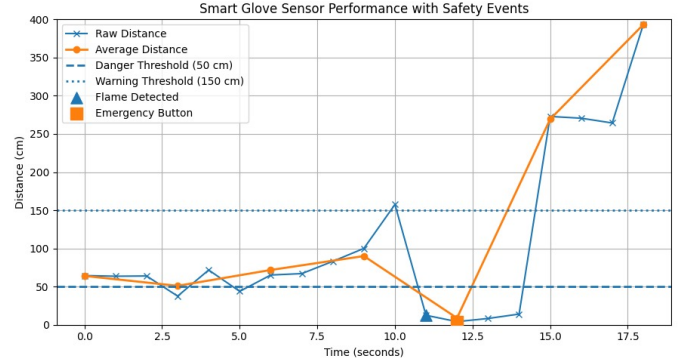


Fig. 4: Ultrasonic sensor distance versus time with safety thresholds and emergency events.

distance values drop drastically as the user moves towards the obstacles in the vicinity and this is the indication of an emergency scenario, which is confirmed by rapid response rate of the system. The steady distance change when the user is in a safe motion testifies to the correct sensor functioning with minimum noise. Based on the measured distance, the system determines the alert level using predefined threshold conditions:

$$\text{Alert Level} = \begin{cases} \text{Safe,} & d > d_{\text{warn}} \\ \text{Warning,} & d_{\text{emg}} < d \leq d_{\text{warn}} \\ \text{Emergency,} & d \leq d_{\text{emg}} \end{cases} \quad (2)$$

The flame sensor was able to detect the sources of fire and high temperatures effectively and therefore gave instant warnings. The sound alert system with the buzzer offered good and quick feedback allowing the users to react without the visual signal and differentiate between the warning and the emergency. The geographical distance between two GPS positions is calculated using the Haversine formula:

$$D = 2R \arcsin \left(\sqrt{\sin^2 \left(\frac{\Delta \phi}{2} \right) + \cos(\phi_1) \cos(\phi_2) \sin^2 \left(\frac{\Delta \lambda}{2} \right)} \right) \quad (3)$$

where R is the Earth's radius, ϕ_1 and ϕ_2 are the latitudes of two positions, and $\Delta \phi$ and $\Delta \lambda$ represent the differences in latitude and longitude, respectively.

To evaluate location tracking and emergency response performance, GPS data were recorded during user movement under normal, warning, and emergency conditions, as illustrated in Figure 5.

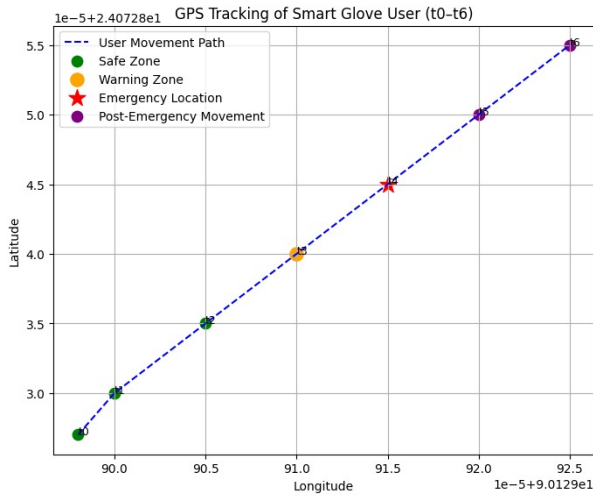


Fig. 5: GPS trajectory of the smart glove user showing safe, warning, emergency, and post-emergency movements.

As shown in Figure 5, the GPS module is able to track the user movement in real time. Distinct segments signify secure movement, caution areas, emergency activation spots, and post emergency motion. The emergency destination is highly identified and this means that the system will be able to provide accurate user location in case of emergency, which is a necessary requirement of providing rapid aid.

TABLE III: Hardware Cost Breakdown of the Proposed Smart Glove

Component	Unit Cost (\$)	Quantity	Total (\$)
Arduino Microcontroller / NodeMCU	9.34	1	9.34
Flame Sensor	0.39	1	0.39
Emergency Button	0.057	1	0.057
Ultrasonic Sensor (HC-SR04)	0.90	1	0.90
GPS Module	3.60	1	3.60
Bluetooth Module	3.11	1	3.11
Buzzer	0.16	1	0.16
Power Supply	0.49	2	0.98
Jumper Wires and Breadboard	1.63	1	1.63
Total			20.167

The GPS module offered good real time location tracking in the outdoor setting with lower accuracy being found in the indoor setting because there are fewer satellites. Table IV gives a synopsis of the quantitative performance of the proposed system. The measured values depict quick reaction, trustworthy barrier recognition and dependable wireless connection, which justifies real time assistive functionality.

Bluetooth communication functioned reliably within its effective range, as summarized in Table IV, allowing stable data exchange between the smart glove and a connected device.

Overall, the experimental results indicate that the proposed smart glove performs effectively as a low-cost assistive wear-

TABLE IV: Performance Summary of the Proposed Smart Glove

Parameter	Observed Value
Obstacle Detection Range	2–400 cm
Warning Threshold	≤ 40 cm
Emergency Detection Distance	< 10 cm
Flame Detection Response	Instant
GPS Accuracy (Outdoor)	$\pm 3\text{--}5$ m
Bluetooth Communication Range	~ 10 m
System Response Time	< 1 s
False Positive Rate	$< 5\%$
Battery Backup Time	5–6 h

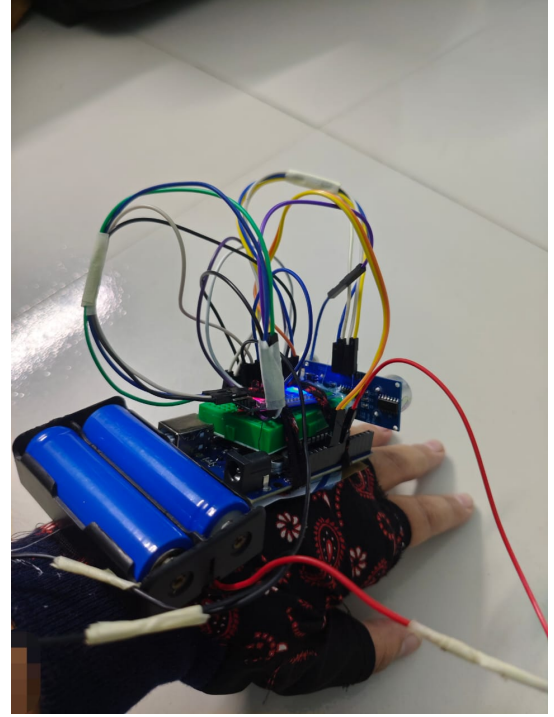


Fig. 6: Prototype Implementation of the Proposed Smart Glove System

able system. While certain limitations were identified under specific environmental conditions, the results confirm the system's potential to improve mobility, safety, and confidence for visually impaired individuals during everyday activities.

VI. DISCUSSION

The designed smart glove reflects the feasibility of IoT-based wearable technology in practice and its use to support visually impaired people in the real world. Obstacle detection, fire hazard, audio warning, and location tracking enables the system to solve various safety and mobility issues at the same time. The findings show that the suggested design can be successfully used to improve the situation awareness and minimize the threat of accidents during daily movement. The ultrasonic sensor worked well in identifying the closest obstacles within the preset field. Nevertheless, object surface texture

and environmental noise were also found to affect detection accuracy and therefore sensor performance may be affected under some circumstances. The fire sensor was a significant addition of security by identifying the presence of fire or high temperature sources. This is a very useful feature especially in buildings and general congested places where one may not be able to see fire hazards at all. Although the sensor has been effective in creating rapid notifications, it can be affected by ambient light and external sources of heat, which means the sensor needs to be carefully adjusted to the threshold. The buzzer-based audio alert system was a very suitable and easy feedback system. Sound alerts allowed users to react fast to the hazards detected without necessarily having to see the information. Nevertheless, audio alerts can be less effective in the noisy environment and it can be recommended to consider adding other types of feedback to increase usability. The addition of GPS and Bluetooth modules allowed real-time tracking and communication of the location, and that is particularly useful in emergency cases. These characteristics enable the caregivers or family members to track the whereabouts of the user and offer assistance on the same. However, GPS positioning may interfere within the indoor space or locations that have weak network coverage. In general, the proposed smart glove offers a low-cost and convenient assistive device capable of enhancing the mobility, safety, and autonomy of visually impaired people. The system combines the detection of obstacles, fire hazard, audio warning, and location tracking features into one wearable gadget, thus tackling several real-life problems users encounter when moving around daily. The design is simple and affordable; this means that it can be used by many people, particularly in resource-restricted environments. Despite the fact that the system has great potential and the basic functionality is reliable, there is still the room to be improved. Future improvements can be made through the application of sophisticated sensor fusion algorithms that can minimize the detection error and false alarms. Additional power consumption and battery optimization would also enhance usability and acceptance by long-term.

VII. CONCLUSION

The paper introduced an IoT-enabled smart glove that incorporates ultrasonic obstacle detection, flame sensing, GPS tracking, and Bluetooth monitoring into a low-priced wearable gadget to enable the visually impaired. Experimental validation demonstrated stability in obstacle detection (within a 2-400 cm range), a reaction time of less than one second, effective flame danger detection, and outdoor GPS resolution of $\pm 3\text{-}5$ m, all at a reasonably affordable price of about \$20.17. The system also fills the key safety gaps in current assistive technologies by integrating real-time hazard detection with sharing emergency location, thereby making them less reliant on caregivers and increasing their independence.

To improve Smart Glove, future work will introduce more sensors (infrared and cameras) to detect obstacles better in complicated settings. A combination of vibration and audio feedback will provide effective notifications in noisy envi-

ronments, whereas voice-based instructions will enable verbal guidance to allow intuitive interaction. Power-saving modes and solar-assisted charging will be employed to increase battery life. Theoretically, lightweight AI algorithms will also be addressed to allow obstacle detection and rudimentary pathfinding and transform the system into a more proactive navigation assistant for the visually impaired user.

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